Supplementary material

How much is the wind worth? Sprints and horizontal jumps from age 10 to senior level

Anonymised for double-anonymous peer review. The tables are numbered in the order of their first citation in the main text.

Supplementary tables

Table S1: Headwind/tailwind asymmetry on the 100 m by sex and age band, against drag-only physics at the athletes' own speeds (slopes in seconds per m/s, signed as gains with tailwind; 95 % CIs clustered on meet).

Sex	Age band	n	Mean time (s)	Tailwind slope	Headwind slope	Head/tail ratio (95 % CI)	Drag-only ratio
Girls/women	12–13	2 733	14.80	0.083	0.112	1.35 (0.82–1.88)	1.81
Girls/women	14–15	9 740	13.93	0.071	0.126	1.76 (1.48–2.04)	1.73
Girls/women	16–17	9 062	13.41	0.059	0.112	1.91 (1.59–2.22)	1.68
Girls/women	18–19	4 207	13.04	0.064	0.106	1.66 (1.31–2.01)	1.66
Girls/women	20–34	4 543	12.57	0.057	0.098	1.71 (1.39–2.03)	1.58
Girls/women	All ages	30 392	13.53	0.065	0.114	1.77 (1.58–1.96)	1.73
Girls/women	All ages, meet FE	30 392	13.53	0.058	0.092	1.59 (1.35–1.83)	1.70
Boys/men	12–13	1 755	14.42	0.071	0.141	1.97 (0.87–3.07)	1.92
Boys/men	14–15	7 499	12.79	0.072	0.113	1.56 (1.28–1.85)	1.73
Boys/men	16–17	9 879	12.01	0.062	0.102	1.65 (1.37–1.93)	1.59
Boys/men	18–19	6 030	11.61	0.067	0.093	1.39 (1.03–1.75)	1.55
Boys/men	20–34	8 909	11.36	0.064	0.082	1.27 (1.01–1.52)	1.58
Boys/men	All ages	34 165	12.08	0.066	0.101	1.54 (1.36–1.71)	1.67
Boys/men	All ages, meet FE	34 165	12.08	0.055	0.080	1.46 (1.18–1.74)	1.63

Head/tail ratio: headwind slope over tailwind slope, CI by the delta method. Drag-only ratio: time changes proportional to the change in squared airspeed at each athlete-season's mean speed, imposed on the observed winds and estimated with the same model (with meet fixed effects in the meet-FE rows). Added after the analysis plan.

Table S2: The elite end with the same estimator: wind effects in US collegiate results (TFRRS) and in the career results of championship finalists (Tilastopaja), with the registry's all-age and senior rows as reference (within athlete × season; 95 % CIs and SEs clustered on meet).

Sample	n	Mean mark	Per m/s, linear (95 % CI)	Per s	Tailwind	Headwind	Head/tail (drag)	0 to +2	Above +2 (SE)	Meet FE
100 m, men										
Registry, all ages	34 165	12.40	0.080 (0.077–0.083)	0.0064	0.066	0.101	1.54 (1.67)	0.073	0.049 (0.005)	0.064
Registry, 20–34	8 909	11.42	0.071 (0.066–0.075)	0.0062	0.064	0.082	1.27 (1.58)	0.073	0.047 (0.007)	0.060
US collegiate (TFRRS)	57 534	10.89	0.054 (0.053–0.056)	0.0050	0.047	0.077	1.62 (1.67)	0.051	0.043 (0.002)	0.045
Championship finalists (Tilastopaja)	8 885	10.19	0.048 (0.045–0.051)	0.0047	0.036	0.070	1.93 (1.45)	0.038	0.033 (0.005)	0.040
100 m, women										
Registry, all ages	30 392	13.84	0.085 (0.082–0.089)	0.0062	0.065	0.114	1.77 (1.73)	0.067	0.059 (0.006)	0.071
Registry, 20–34	4 543	12.63	0.072 (0.067–0.078)	0.0057	0.057	0.098	1.71 (1.58)	0.055	0.063 (0.008)	0.062
US collegiate (TFRRS)	63 234	12.22	0.058 (0.056–0.060)	0.0047	0.050	0.081	1.61 (1.76)	0.059	0.041 (0.002)	0.047
Championship finalists (Tilastopaja)	8 047	11.25	0.059 (0.055–0.063)	0.0052	0.044	0.083	1.86 (1.49)	0.047	0.039 (0.008)	0.053
200 m, men										
Registry, all ages	29 352	27.07	0.117 (0.108–0.127)	0.0043	0.071	0.183	2.56 (1.82)	0.081	0.053 (0.015)	0.067
Registry, 20–34	4 931	23.08	0.085 (0.068–0.101)	0.0037	0.048	0.151	3.15 (1.64)	0.070	0.013 (0.027)	0.053
US collegiate (TFRRS)	57 111	22.05	0.066 (0.062–0.069)	0.0030	0.052	0.104	2.00 (1.64)	0.057	0.045 (0.005)	0.050
Championship finalists (Tilastopaja)	6 265	20.54	0.065 (0.057–0.074)	0.0032	0.030	0.114	3.74 (1.41)	0.034	0.021 (0.016)	0.068
200 m, women										
Registry, all ages	29 894	29.86	0.125 (0.115–0.136)	0.0042	0.083	0.185	2.22 (1.87)	0.110	0.033 (0.015)	0.077
Registry, 20–34	2 954	25.83	0.095 (0.075–0.115)	0.0037	0.047	0.180	3.81 (1.76)	0.071	0.011 (0.029)	0.059
US collegiate (TFRRS)	71 467	25.30	0.074 (0.070–0.078)	0.0029	0.057	0.118	2.06 (1.81)	0.072	0.040 (0.005)	0.056
Championship finalists (Tilastopaja)	5 804	22.93	0.078 (0.069–0.088)	0.0034	0.055	0.110	2.00 (1.50)	0.063	0.037 (0.022)	0.079
100 m hurdles, women										
US collegiate (TFRRS)	61 051	14.74	0.075 (0.072–0.079)	0.0051	0.057	0.129	2.27	0.072	0.038 (0.005)	0.062
Championship finalists (Tilastopaja)	7 302	12.94	0.054 (0.049–0.059)	0.0041	0.033	0.084	2.51	0.031	0.039 (0.010)	0.045
110 m hurdles, men										
US collegiate (TFRRS)	45 225	14.97	0.057 (0.054–0.060)	0.0038	0.043	0.099	2.31	0.051	0.034 (0.004)	0.043
Championship finalists (Tilastopaja)	6 788	13.47	0.044 (0.039–0.050)	0.0033	0.026	0.072	2.77	0.027	0.023 (0.009)	0.046
Long jump, men										
Registry, all ages	15 259	4.69	3.34 (2.90–3.78)	—	2.98	4.27	1.43	1.54	5.73 (0.60)	4.77
Registry, 20–34	1 813	6.26	3.11 (1.69–4.52)	—	2.85	4.17	1.46	2.70	3.11 (1.71)	8.56
US collegiate (TFRRS)	38 335	6.77	3.74 (3.46–4.02)	—	3.88	3.18	0.82	4.97	2.58 (0.31)	3.35
Championship finalists (Tilastopaja)	6 279	7.95	4.53 (3.95–5.11)	—	4.08	6.42	1.57	4.89	2.40 (0.94)	4.13
Long jump, women										
Registry, all ages	20 515	4.17	3.04 (2.70–3.38)	—	2.88	3.40	1.18	2.51	3.58 (0.43)	4.18
Registry, 20–34	1 536	5.43	4.15 (3.03–5.27)	—	3.99	4.88	1.22	4.38	3.42 (1.26)	7.20
US collegiate (TFRRS)	50 068	5.44	2.73 (2.52–2.94)	—	2.86	2.25	0.79	3.81	1.65 (0.23)	2.37
Championship finalists (Tilastopaja)	6 290	6.58	3.71 (3.27–4.16)	—	3.41	4.96	1.46	4.62	1.04 (0.58)	2.84
Triple jump, men										
Registry, all ages	4 237	11.15	4.21 (3.14–5.28)	—	5.18	1.27	0.25	3.39	8.50 (1.60)	7.39
Registry, 20–34	527	13.66	4.89 (2.55–7.23)	—	7.68	−5.28	-0.69	3.94	14.51 (4.48)	10.77
US collegiate (TFRRS)	24 224	14.25	4.57 (4.15–5.00)	—	4.27	5.76	1.35	5.54	2.68 (0.57)	3.56
Championship finalists (Tilastopaja)	4 801	16.85	6.55 (5.45–7.65)	—	5.74	9.50	1.65	5.83	5.54 (1.46)	5.08
Triple jump, women										
Registry, all ages	7 421	9.89	4.15 (3.41–4.90)	—	4.30	3.70	0.86	3.79	5.28 (1.14)	6.02
Registry, 20–34	1 018	11.71	4.43 (3.14–5.72)	—	5.16	0.49	0.10	5.91	3.86 (1.80)	6.30
US collegiate (TFRRS)	32 378	11.68	3.94 (3.64–4.25)	—	3.78	4.60	1.22	4.90	2.34 (0.39)	2.88
Championship finalists (Tilastopaja)	4 467	14.21	5.89 (5.09–6.68)	—	5.86	6.00	1.02	6.88	3.66 (1.22)	5.28

Sprints in seconds, jumps in centimetres, per m/s of wind, signed as gains with tailwind; n is the number of results in athlete-season-event cells with at least two wind-measured marks. Mean mark: all analysed results (s for sprints, m for jumps), as in Table S4. Per s: the linear gain divided by the mean time. The registry's 20–34 jump rows rest on thin samples. Tailwind and headwind: the separate branch slopes; head/tail: their ratio, with in parentheses the ratio drag-only physics predicts for the same winds and estimator at the athletes' own speeds (100 m and 200 m). 0 to +2 m/s and above +2 m/s: the tailwind slopes of the piecewise model used for H5. Meet FE: the linear slope with month dummies replaced by meet fixed effects. The registry's hurdle events are class-specific (Table 5), so no registry row is given for the hurdles. The external estimates were stable across the registry's robustness variants (cells with at least three results, no exact-zero readings, athlete clustering; for the collegiate sample also without the 2016–2017 seasons, whose short careers the collection under-covers): the 100 and 200 m slopes moved by at most 3 %, and meet fixed effects attenuated the 100 m slopes by 10–20 % in all three samples. External samples: all outdoor results in the six events in the TFRRS profiles of NCAA Division I athletes (shared cleaned build, version 1.2), and the career results of athletes who reached a World Championships, Olympic or other major championship final 1983–2026 (Tilastopaja), after the registry's filters (electronic timing, wind within ±8 m/s, event-specific plausibility bounds; combined-event sub-results excluded). Added after the analysis plan.

Table S3: Robustness: gain per +1 m/s under alternative specifications (point estimates; s for sprints, cm for jumps).

Event	Primary	Cells ≥3	No exact 0.0	Excl. ±0.2 of +2.0	2011–19	2021–26	Meet FE
60 m	0.0591	0.0590	0.0591	0.0587	0.0600	0.0573	0.0513
80 m	0.0746	0.0711	0.0761	0.0729	0.0740	0.0779	0.0768
100 m	0.0823	0.0816	0.0823	0.0829	0.0843	0.0795	0.0659
150 m	0.0814	—	0.0780	0.0850	—	—	0.0838
200 m	0.1215	0.1203	0.1215	0.1212	0.1258	0.1135	0.0713
Long jump	3.1597	3.1129	3.2467	3.1932	3.2987	2.9343	4.1017
Long jump (zone)	2.1133	2.0471	2.1538	2.0984	2.1337	2.1518	2.6624
Triple jump	4.1848	4.1647	4.2642	4.1691	4.0189	4.7732	6.1704
Triple jump (zone)	3.8821	3.3154	3.8589	3.7725	3.7075	4.8247	5.0930

Point estimates are identical under athlete and meet clustering; only intervals differ. The 150 m lacks the cells-≥3 and period columns for want of data.

Table S4: Effect per second of race time: gain per +1 m/s divided by mean race duration, flat sprints and short hurdles.

Event	Mean time (s)	Gain per +1 m/s (s)	Per second of race time
60 m	9.67	0.059	0.00611
80 m	12.58	0.075	0.00594
100 m	13.09	0.082	0.00629
150 m	20.22	0.081	0.00403
200 m	28.47	0.121	0.00427
60 m hurdles 68 cm	13.16	0.063	0.00478
60 m hurdles 76.2 cm	11.82	0.062	0.00525
80 m hurdles 76.2 cm	13.36	0.072	0.00535
80 m hurdles 84 cm	14.09	0.078	0.00551
100 m hurdles 76.2 cm	15.61	0.108	0.00694
100 m hurdles 84 cm	14.94	0.078	0.00523
100 m hurdles 91.4 cm	15.66	0.084	0.00534
110 m hurdles 91.4 cm	15.90	0.103	0.00646
110 m hurdles 100 cm	15.74	0.096	0.00608
110 m hurdles 106.7 cm	15.53	0.084	0.00540

Straight flat sprints cluster at 0.0059–0.0063 per m/s per second of race time; the bend events (150/200 m) sit about a third lower; hurdle codes bracket the flat rates.

Table S5: Wind effect in the horizontal jumps by age band (gain per +1 m/s; values girls/women / boys/men; 95 % CIs clustered on meet).

Event	Unit	10–11	12–13	14–15	16–17	18–19	20–34
Long jump	cm	—	—	2.89 (2.42–3.35) / 3.36 (2.82–3.89)	3.04 (2.48–3.60) / 3.71 (2.68–4.74)	3.20 (2.40–3.99) / 2.94 (1.58–4.29)	4.15 (3.03–5.27) / 3.11 (1.69–4.52)
Long jump	%	—	—	0.70 (0.58–0.82) / 0.76 (0.62–0.90)	0.62 (0.49–0.76) / 0.66 (0.44–0.89)	0.64 (0.46–0.82) / 0.48 (0.23–0.73)	0.76 (0.53–1.00) / 0.49 (0.23–0.75)
Long jump (zone)	cm	1.83 (1.41–2.24) / 2.39 (1.88–2.90)	1.86 (1.59–2.14) / 2.54 (2.17–2.91)	—	—	—	—
Long jump (zone)	%	0.55 (0.42–0.68) / 0.70 (0.54–0.85)	0.49 (0.41–0.56) / 0.66 (0.56–0.75)	—	—	—	—
Triple jump	cm	—	—	4.18 (3.07–5.29) / 4.20 (2.49–5.91)	4.13 (2.88–5.37) / 4.10 (2.09–6.11)	3.52 (1.68–5.36) / 4.21 (1.67–6.76)	4.43 (3.14–5.72) / 4.89 (2.55–7.23)
Triple jump	%	—	—	0.45 (0.33–0.57) / 0.41 (0.24–0.58)	0.40 (0.28–0.52) / 0.32 (0.15–0.49)	0.31 (0.13–0.49) / 0.32 (0.12–0.52)	0.37 (0.26–0.48) / 0.34 (0.17–0.51)

In each cell the first value is girls/women, the second boys/men. The zone events are contested only in the youngest classes. In centimetres the effect rises with age for girls and women as jump distances and flight times grow; in percent it shows no consistent age direction.

Table S6: Quadratic specification: linear and quadratic wind terms (raw outcome units per m/s: seconds for sprints, metres for jumps, e.g. 0.0328 m = 3.28 cm; SEs clustered on meet).

Event	Linear term (SE)	Quadratic term (SE)	n
60 m	−0.0596 (0.0010)	0.0037 (0.0004)	72 558
80 m	−0.0752 (0.0065)	0.0013 (0.0026)	1 410
100 m	−0.0856 (0.0014)	0.0057 (0.0005)	64 557
150 m	−0.0794 (0.0194)	−0.0052 (0.0109)	328
200 m	−0.1307 (0.0038)	0.0141 (0.0013)	59 246
Long jump	0.0328 (0.0023)	−0.0008 (0.0008)	35 774
Long jump (zone)	0.0194 (0.0012)	0.0015 (0.0004)	44 336
Triple jump	0.0382 (0.0047)	0.0020 (0.0016)	11 658
Triple jump (zone)	0.0342 (0.0060)	0.0030 (0.0018)	5 426

Negative linear terms for sprints mean faster times with tailwind. Positive quadratic terms in the large sprints match the headwind-tailwind asymmetry of Table 4.

Table S7: Correction table: tailwind slope over 0 to +2 m/s, and separate tail- and headwind slopes, by event, sex and age band (sprints; jumps event-level, in cm), with 95 % CIs clustered on meet.

Event	Sex	Age band	Tailwind 0 to +2	Tailwind slope per +1 m/s (95 % CI)	Headwind slope per −1 m/s (95 % CI)	n
60 m	Boys/men	10–11	0.057	0.055 (0.046–0.064)	0.088 (0.077–0.098)	7 635
60 m	Boys/men	12–13	0.066	0.056 (0.050–0.062)	0.071 (0.064–0.078)	15 457
60 m	Boys/men	14–15	0.054	0.042 (0.036–0.049)	0.062 (0.051–0.073)	6 955
60 m	Boys/men	16–17	0.046	0.019 (0.004–0.035)	0.064 (0.049–0.080)	570
60 m	Girls/women	10–11	0.055	0.044 (0.036–0.052)	0.085 (0.076–0.094)	9 794
60 m	Girls/women	12–13	0.057	0.047 (0.042–0.051)	0.067 (0.062–0.072)	21 886
60 m	Girls/women	14–15	0.046	0.038 (0.032–0.043)	0.068 (0.062–0.075)	9 320
60 m	Girls/women	16–17	0.030	0.021 (0.007–0.034)	0.091 (0.068–0.113)	389
100 m	Boys/men	12–13	0.082	0.071 (0.041–0.102)	0.141 (0.111–0.170)	1 755
100 m	Boys/men	14–15	0.073	0.072 (0.063–0.081)	0.113 (0.102–0.123)	7 499
100 m	Boys/men	16–17	0.069	0.062 (0.054–0.070)	0.102 (0.094–0.111)	9 879
100 m	Boys/men	18–19	0.079	0.067 (0.058–0.077)	0.093 (0.080–0.107)	6 030
100 m	Boys/men	20–34	0.073	0.064 (0.058–0.071)	0.082 (0.071–0.092)	8 909
100 m	Girls/women	12–13	0.073	0.083 (0.062–0.105)	0.112 (0.091–0.134)	2 733
100 m	Girls/women	14–15	0.073	0.071 (0.063–0.079)	0.126 (0.116–0.135)	9 740
100 m	Girls/women	16–17	0.066	0.059 (0.052–0.066)	0.112 (0.102–0.122)	9 062
100 m	Girls/women	18–19	0.072	0.064 (0.057–0.072)	0.106 (0.092–0.120)	4 207
100 m	Girls/women	20–34	0.055	0.057 (0.050–0.064)	0.098 (0.087–0.109)	4 543
200 m	Boys/men	10–11	0.133	0.107 (0.023–0.191)	0.194 (0.090–0.298)	1 301
200 m	Boys/men	12–13	0.081	0.076 (0.041–0.111)	0.217 (0.174–0.259)	5 443
200 m	Boys/men	14–15	0.079	0.078 (0.059–0.096)	0.188 (0.164–0.212)	7 147
200 m	Boys/men	16–17	0.094	0.074 (0.060–0.088)	0.172 (0.146–0.198)	6 372
200 m	Boys/men	18–19	0.063	0.069 (0.053–0.084)	0.149 (0.124–0.173)	4 158
200 m	Boys/men	20–34	0.070	0.048 (0.026–0.070)	0.151 (0.125–0.178)	4 931
200 m	Girls/women	10–11	0.256	0.099 (0.025–0.173)	0.086 (−0.012–0.185)	1 780
200 m	Girls/women	12–13	0.127	0.105 (0.080–0.130)	0.181 (0.143–0.219)	7 710
200 m	Girls/women	14–15	0.100	0.094 (0.075–0.113)	0.186 (0.162–0.211)	8 640
200 m	Girls/women	16–17	0.091	0.066 (0.049–0.082)	0.208 (0.173–0.243)	5 854
200 m	Girls/women	18–19	0.081	0.065 (0.047–0.082)	0.151 (0.120–0.183)	2 956
200 m	Girls/women	20–34	0.071	0.047 (0.025–0.070)	0.180 (0.143–0.218)	2 954
Long jump	Both	all	—	2.93 (2.58–3.28) cm	3.71 (2.77–4.65) cm	35 774
Long jump (zone)	Both	all	—	2.48 (2.19–2.77) cm	1.41 (0.95–1.87) cm	44 336
Triple jump	Both	all	—	4.63 (3.78–5.48) cm	2.79 (0.78–4.80) cm	11 658
Triple jump (zone)	Both	all	—	4.69 (3.40–5.98) cm	1.90 (−0.25–4.04) cm	5 426

Worked example: a girls' 12–13 100 m run in +1.8 m/s is corrected by $1.8 \times 0.073 = 0.13$ s to still air, using the slope over 0 to +2 m/s. Sprint slopes are in seconds; the four jump rows are in centimetres, pooled across bands. The 60 m rows at 16–17 rest on thin outdoor samples.

Table S8: Meet fixed effects by age band: effect per m/s of wind with month dummies replaced by meet fixed effects (seconds, 95 % CI clustered on meet), sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.055 (0.045–0.066)	0.055 (0.049–0.061)	0.047 (0.040–0.055)	0.030 (0.011–0.048)	—	—
60 m	Girls/women	0.056 (0.049–0.063)	0.049 (0.043–0.054)	0.047 (0.041–0.052)	0.030 (0.012–0.048)	—	—
100 m	Boys/men	—	0.107 (0.072–0.143)	0.071 (0.056–0.085)	0.063 (0.050–0.075)	0.060 (0.046–0.073)	0.060 (0.051–0.069)
100 m	Girls/women	—	0.093 (0.077–0.109)	0.077 (0.066–0.089)	0.065 (0.057–0.072)	0.069 (0.059–0.080)	0.062 (0.054–0.070)
200 m	Boys/men	−0.000 (−0.147–0.146)	0.091 (0.054–0.128)	0.078 (0.054–0.102)	0.063 (0.038–0.088)	0.058 (0.026–0.090)	0.053 (0.033–0.072)
200 m	Girls/women	0.030 (−0.082–0.142)	0.087 (0.060–0.113)	0.080 (0.061–0.100)	0.079 (0.057–0.100)	0.071 (0.047–0.094)	0.059 (0.006–0.111)

Same reporting threshold as Table 3. The 200 m at 10–11 has too few results per meet for the meet-FE estimate to be informative.